THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 220 cm²	$2 \times \frac{22}{7} \times 7 \times 5$; CSA $= 220$ cm^2
2	**(b)** 15	$\frac{10+20}{2}$; Class mark $= 15$
3	**(a)** $\frac{1}{2}$	Favourable outcomes: 2, 4, 6; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
4	**(c)** 1540 cm³	$\frac{22}{7} \times 7^2 \times 10$; Volume $= 1540$ cm^3
5	**(b)** 30–40	It belongs to the class 30–40; Modal class $=$ 30–40
6	**(d)** $\frac{1}{3}$	Favourable outcomes: 3, 6; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
7	**(a)** 25 cm	$l^2 = 7^2 + 24^2 = 625$; $l = 25$ cm
8	**(b)** 40–50	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class $=$ 40–50
9	**(d)** $\frac{1}{4}$	Favourable outcomes: HH; number $= 1$; $P = \frac{1}{4}$
10	**(b)** 550 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 7 \times 25 = 550$ cm^2
11	**(d)** 50	$3 + 9 + 4 + 18 + 16$; $\sum f_i = 50$
12	**(c)** $\frac{3}{8}$	Favourable balls $= 3$; $P = \frac{3}{8}$
13	**(d)** 616 cm²	$4 \times \frac{22}{7} \times 7^2$; Surface area $= 616$ cm^2
14	**(c)** 12	$3 + 9$; $cf = 12$
15	**(d)** $\frac{3}{10}$	Favourable numbers: 3, 6, 9; number $= 3$; $P = \frac{3}{10}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Total surface area $= 2\pi r(r + h)$; $2 \times \frac{22}{7} \times 7 \times (7 + 10)$ (1); TSA $= 748$ cm^2 (1).	1+1
17	Class marks x_i: 15, 25, 35, 45; $\sum f_i = 50$ and $\sum f_i x_i = 1740$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1740}{50} = 34.8$ (1).	1+1
18	Total possible outcomes $= 6$: 1, 2, 3, 4, 5, 6; Favourable outcomes: ; number $= 0$ (1); $P = \frac{0}{6} = 0$ (1).	1+1
19	Volume $= \frac{2}{3}\pi r^3$; $\frac{2}{3} \times \frac{22}{7} \times 10.5^3$ (1); Volume $= 2425.5$ cm^3 (1).	1+1
20	Highest frequency is 20, so the modal class is 50–60; $l = 50$, $f_1 = 20$, $f_0 = 10$, $f_2 = 5$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{10}{25} \times 10 = 54$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 7^2 \times 24$ (1); Volume $= 1232$ cm^3 (1).	1+1+1
22	Class marks x_i: 5, 15, 25, 35, 45 (1); $\sum f_i = 50$ and $\sum f_i x_i = 970$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{970}{50} = 19.4$ (1).	1+1+1
23	Total tickets $= 10 + 25 = 35$ (1); Favourable tickets (prizes) $= 10$ (1); $P = \frac{10}{35} = \frac{2}{7}$ (1).	1+1+1
24	Slant height $l = 25$ cm (1); TSA $= \pi r(l + r)$ (1); TSA $= \frac{22}{7} \times 7 \times (25 + 7) = 704$ cm^2 (1).	1+1+1
	OR: Cumulative frequencies: 5, 21, 26, 39, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 40–50: $l = 40$, $cf = 21$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{25-21}{5} \times 10 = 48$ (1).	1+1+1
25	Total pens $= 144$; good pens $= 144 - 20 = 124$ (1); Favourable outcomes $=$ number of good pens (1); $P = \frac{124}{144} = \frac{31}{36}$ (1).	1+1+1
	OR: TSA $= 6a^2$ (1); $a^2 = 9$, so $a = 3$ cm (1); Volume $= a^3 = 27$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24 \times 12 \times 6}{3^3}$ (2); $n = 64$ (1).	2+2+1
	OR: Take assumed mean $a = 85$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (2); $\sum f_i = 50$ and $\sum f_i d_i = -510$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 85 + \frac{-510}{50} = 74.8$ (1).	2+2+1

27	Total balls $= 10$ (2); Favourable balls: 3, 6, 0 (2); Probabilities: $\frac{3}{10}$, $\frac{3}{5}$, 0 (1).	2+2+1
	OR: Volume of big sphere $= n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(6)^3}{\frac{4}{3}\pi(2)^3} = \left(\frac{6}{2}\right)^3$ (2); $n = 27$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 38808 cm³	$\frac{4}{3} \times \frac{22}{7} \times 21^3$; Volume $= 38808$ cm^3
2	**(b)** 34.8	$\sum f_i = 50$ and $\sum f_i x_i = 1740$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1740}{50} = 34.8$
3	**(d)** $\frac{3}{4}$	$P(\text{not } E) = 1 - \frac{1}{4}$; $P(\text{not } E) = \frac{3}{4}$
4	**(a)** 462 cm²	$3 \times \frac{22}{7} \times 7^2$; TSA $= 462$ cm^2
5	**(d)** 58.75	$l = 50$, $f_1 = 18$, $f_0 = 4$, $f_2 = 16$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 50 + \frac{14}{16} \times 10 = 58.75$
6	**(d)** 1	A sure event always happens
7	**(a)** 880 cm²	$2 \times \frac{22}{7} \times 14 \times 10$; CSA $= 880$ cm^2
8	**(c)** 21	Median class 20–30: $l = 20$, $cf = 12$, $f = 5$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{12.5 - 12}{5} \times 10 = 21$
9	**(a)** $\frac{1}{2}$	Favourable outcomes: 1, 3, 5; number $= 3$; $P = \frac{3}{6} = \frac{1}{2}$
10	**(c)** 3080 cm³	$\frac{22}{7} \times 14^2 \times 5$; Volume $= 3080$ cm^3
11	**(a)** 30	Class limits
12	**(d)** $\frac{1}{3}$	Favourable outcomes: 1, 4; number $= 2$; $P = \frac{2}{6} = \frac{1}{3}$
13	**(a)** 50 cm	$l^2 = 14^2 + 48^2 = 2500$; $l = 50$ cm
14	**(b)** 25	$\frac{20+30}{2}$; Class mark $= 25$
15	**(c)** $\frac{3}{4}$	Favourable outcomes: HH, HT, TH; number $= 3$; $P = \frac{3}{4}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Total outcomes $= 2^2 = 4$: HH, HT, TH, TT; Favourable outcomes: HH; number $= 1$ (1); $P = \frac{1}{4}$ (1).	1+1
17	TSA $= 2(lb + bh + hl)$; $2(8 \times 5 + 5 \times 3 + 3 \times 8)$ (1); TSA $= 158$ cm^2 (1).	1+1
18	Cumulative frequencies: 4, 14, 18, 25; $N = 25$, so $\frac{N}{2} = 12.5$; Median class 50–60: $l = 50$, $cf = 4$, $f = 10$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 50 + \frac{12.5 - 4}{10} \times 10 = 58.5$ (1).	1+1
19	Total balls $= 5 + 3 = 8$; Favourable balls $= 3$ (1); $P = \frac{3}{8}$ (1).	1+1
20	Total surface area $= 2\pi r(r + h)$; $2 \times \frac{22}{7} \times 14 \times (14 + 6)$ (1); TSA $= 1760$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Highest frequency is 10, so the modal class is 20–30 (1); $l = 20$, $f_1 = 10$, $f_0 = 2$, $f_2 = 2$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{8}{16} \times 10 = 25$ (1).	1+1+1
22	Total students $= 40$; boys $= 40 - 24 = 16$ (1); Favourable outcomes $= 16$ (1); $P = \frac{16}{40} = \frac{2}{5}$ (1).	1+1+1
23	Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 14^2 \times 48$ (1); Volume $= 9856$ cm^3 (1).	1+1+1
24	Class marks x_i: 5, 15, 25, 35, 45 (1); $\sum f_i = 50$ and $\sum f_i x_i = 1240$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1240}{50} = 24.8$ (1).	1+1+1
	OR: Total tickets $= 8 + 32 = 40$ (1); Favourable tickets (prizes) $= 8$ (1); $P = \frac{8}{40} = \frac{1}{5}$ (1).	1+1+1
25	Slant height $l = 50$ cm (1); TSA $= \pi r(l + r)$ (1); TSA $= \frac{22}{7} \times 14 \times (50 + 14) = 2816$ cm^2 (1).	1+1+1
	OR: Cumulative frequencies: 5, 22, 27, 45, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 70–80: $l = 70$, $cf = 22$, $f = 5$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{25 - 22}{5} \times 10 = 76$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Cumulative frequencies: 10, 12, 18, 34, 40; $N = 40$, so $\frac{N}{2} = 20$ (2); Median class 40–50: $l = 40$, $cf = 18$, $f = 16$, $h = 10$ (2); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{20 - 18}{16} \times 10 = 41.25$ (1).	2+2+1

	OR: Total outcomes $= 12$ (2); Favourable counts: 9, 5, 0 (2); Probabilities: $\frac{3}{4}$, $\frac{5}{12}$, 0 (1).	2+2+1
27	Volume of cuboid $= n\times$ volume of a cube (2); $n = \frac{24\times12\times12}{6^3}$ (2); $n = 16$ (1).	2+2+1
	OR: Take assumed mean $a = 35$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (2); $\sum f_i = 40$ and $\sum f_i d_i = 0$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 35 + \frac{0}{40} = 35$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 2200 cm²	CSA $= \pi rl$; CSA $= \frac{22}{7} \times 14 \times 50 = 2200$ cm^2
2	**(c)** 10–20	It belongs to the class 10–20; Modal class = 10–20
3	**(b)** $\frac{3}{5}$	Favourable balls = 6; $P = \frac{6}{10} = \frac{3}{5}$
4	**(c)** 2464 cm²	$4 \times \frac{22}{7} \times 14^2$; Surface area = 2464 cm^2
5	**(c)** 60–70	The median class is the class whose cumulative frequency first exceeds $\frac{N}{2}$; Median class = 60–70
6	**(a)** $\frac{5}{12}$	Favourable numbers: 2, 3, 5, 7, 11; number = 5; $P = \frac{5}{12}$
7	**(c)** 310464 cm³	$\frac{4}{3} \times \frac{22}{7} \times 42^3$; Volume = 310464 cm^3
8	**(a)** 25	$3 + 5 + 9 + 8$; $\sum f_i = 25$
9	**(a)** $\frac{4}{7}$	$P(\text{not } E) = 1 - \frac{3}{7}$; $P(\text{not } E) = \frac{4}{7}$
10	**(d)** 1848 cm²	$3 \times \frac{22}{7} \times 14^2$; TSA = 1848 cm^2
11	**(b)** 45	$17 + 20 + 8$; $cf = 45$
12	**(b)** 0	It never happens
13	**(d)** 1056 cm²	$2 \times \frac{22}{7} \times 21 \times 8$; CSA = 1056 cm^2
14	**(d)** 16.6	$\sum f_i = 25$ and $\sum f_i x_i = 415$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{415}{25} = 16.6$
15	**(a)** $\frac{1}{3}$	Favourable outcomes: 1, 2; number = 2; $P = \frac{2}{6} = \frac{1}{3}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Class marks x_i: 5, 15, 25, 35; $\sum f_i = 25$ and $\sum f_i x_i = 415$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{415}{25} = 16.6$ (1).	1+1
17	Total possible outcomes = 6: 1, 2, 3, 4, 5, 6; Favourable outcomes: 5, 6; number = 2 (1); $P = \frac{2}{6} = \frac{1}{3}$ (1).	1+1
18	Volume $= \frac{2}{3}\pi r^3$; $\frac{2}{3} \times \frac{22}{7} \times 21^3$ (1); Volume = 19404 cm^3 (1).	1+1
19	Highest frequency is 20, so the modal class is 40–50; $l = 40$, $f_1 = 20$, $f_0 = 17$, $f_2 = 3$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \frac{3}{20} \times 10 = 41.5$ (1).	1+1
20	Total outcomes $= 2^2 = 4$: HH, HT, TH, TT; Favourable outcomes: HH, HT, TH; number = 3 (1); $P = \frac{3}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Total pens = 100; good pens = $100 - 12 = 88$ (1); Favourable outcomes = number of good pens (1); $P = \frac{88}{100} = \frac{22}{25}$ (1).	1+1+1
22	TSA $= 6a^2$ (1); $a^2 = 16$, so $a = 4$ cm (1); Volume $= a^3 = 64$ cm^3 (1).	1+1+1
23	Highest frequency is 15, so the modal class is 20–30 (1); $l = 20$, $f_1 = 15$, $f_0 = 11$, $f_2 = 9$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{4}{10} \times 10 = 24$ (1).	1+1+1
24	Total students = 30; boys $= 30 - 12 = 18$ (1); Favourable outcomes = 18 (1); $P = \frac{18}{30} = \frac{3}{5}$ (1).	1+1+1
	OR: Volume $= \frac{1}{3}\pi r^2 h$ (1); $\frac{1}{3} \times \frac{22}{7} \times 21^2 \times 28$ (1); Volume = 12936 cm^3 (1).	1+1+1
25	Class marks x_i: 25, 35, 45, 55, 65 (1); $\sum f_i = 50$ and $\sum f_i x_i = 2100$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2100}{50} = 42$ (1).	1+1+1
	OR: Total tickets $= 5 + 45 = 50$ (1); Favourable tickets (prizes) = 5 (1); $P = \frac{5}{50} = \frac{1}{10}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Total balls = 20 (2); Favourable balls: 8, 12, 0 (2); Probabilities: $\frac{2}{5}$, $\frac{3}{5}$, 0 (1).	2+2+1
	OR: Volume of big sphere = $n\times$ volume of a small sphere (2); $n = \frac{\frac{4}{3}\pi(9)^3}{\frac{4}{3}\pi(3)^3} = \left(\frac{9}{3}\right)^3$ (2); $n = 27$ (1).	2+2+1

27	Cumulative frequencies: 11, 13, 23, 36, 40; $N = 40$, so $\frac{N}{2} = 20$ (2); Median class 70–80: $l = 70$, $cf = 13$, $f = 10$, $h = 10$ (2); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{20-13}{10} \times 10 = 77$ (1).	2+2+1
	OR: Total outcomes $= 20$ (2); Favourable counts: 6, 8, 5 (2); Probabilities: $\frac{3}{10}$, $\frac{2}{5}$, $\frac{1}{4}$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND